

WASP-10b

R-1710

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_190901 · created 2026-08-02T15:29:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458727.8775 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.058
Transit depth (Rp/Rs)^2	1.36 +/- 1.36 [%]
Orbital Inclination (inc)	84.07 +/- 1.93
Airmass coefficient 1 (a1)	0.215 +/- 0.098
Airmass coefficient 2 (a2)	1.32 +/- 0.47
Scatter in the residuals of the lightcurve fit is	40.18 %
Best Comparison Star	None
Optimal Aperture	1.58
Optimal Annulus	5.0
Transit Duration (day)	0.024 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.058 vs 0.1592 (deviation 0.56σ)
Duration fit vs published	0.024 d vs 0.0946 d (deviation 1.76σ)
Mid-time O–C	-7.5 min
Depth SNR	1.6
Residual scatter	40.18 %

Light curve

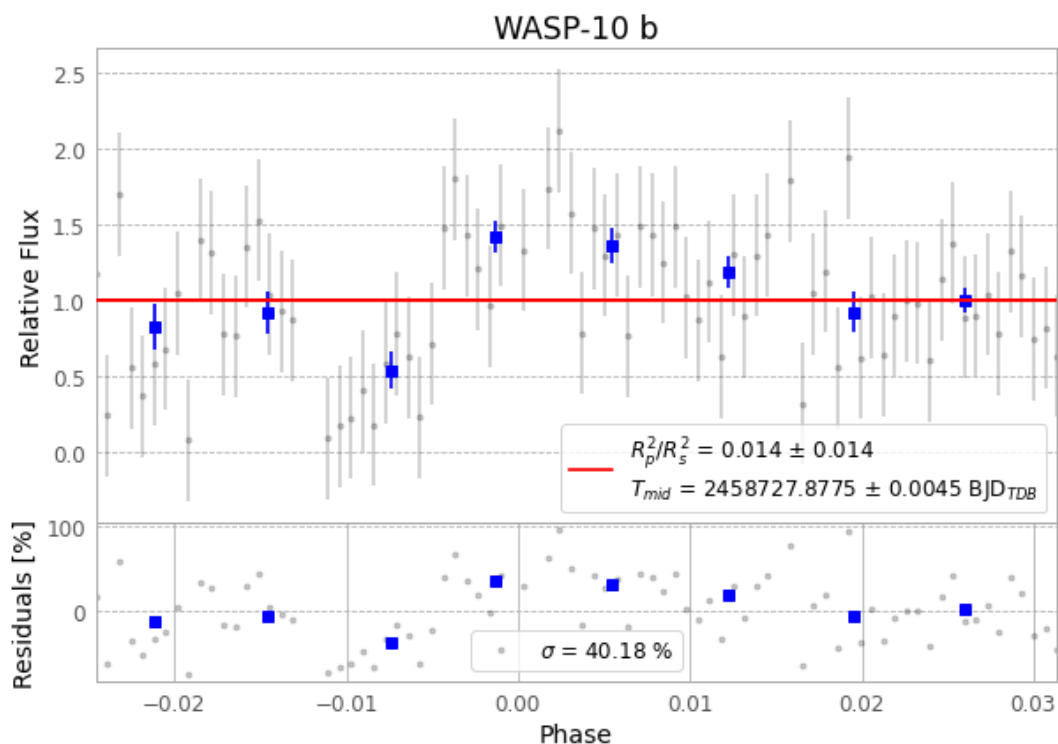


Figure 1 — FinalLightCurve_WASP-10 b_2019-09-01.png (SHA-256 144b17c3d21e42b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [491, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 51537647e88025a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-10190901070612.FITS → WASP-10190901111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-190901.log (05b946038a30...), FinalLightCurve_WASP-10 b_2019-09-01.png (144b17c3d21e...), FinalLightCurve_WASP-10 b_2019-09-01.csv (03312c8baaca...)

Compute resources

Wall time 270.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 15:29Z.